

HW12

Zeyu Qi (A17342618)

Q13: Read this file into R and determine the sample size for each genotype and their corresponding median expression levels for each of these genotypes

```
expr <- read.table("rs8067378_ENSG00000172057.6.txt")
head(expr)
```

	sample	geno	exp
1	HG00367	A/G	28.96038
2	NA20768	A/G	20.24449
3	HG00361	A/A	31.32628
4	HG00135	A/A	34.11169
5	NA18870	G/G	18.25141
6	NA11993	A/A	32.89721

```
nrow(expr)
```

```
[1] 462
```

```
table(expr$geno)
```

```
A/A A/G G/G
108 233 121
```

```
aggregate(exp ~ geno, data = expr, FUN = function(x) c(
  sample_size = length(x),
  median_expression = median(x, na.rm = TRUE)
))
```

	geno	exp.sample_size	exp.median_expression
1	A/A	108.00000	31.24847
2	A/G	233.00000	25.06486
3	G/G	121.00000	20.07363

```
aggregate(exp ~ geno, data = expr, FUN = median)
```

	geno	exp
1	A/A	31.24847
2	A/G	25.06486
3	G/G	20.07363

Individuals with genotype A/A have a sample size of 108 samples and the highest median expression level: 31.24847

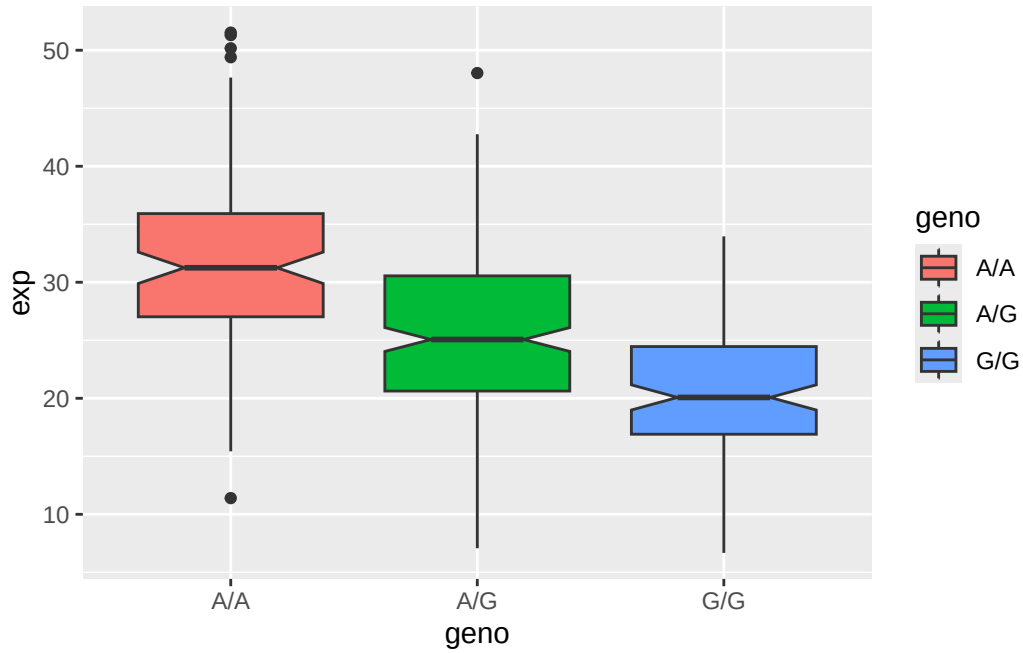
Individuals with genotype A/G have a sample size of 233 samples and an intermediate median expression level: 25.06486

Individuals with genotype G/G have a sample size of 121 samples and the lowest median expression level: 20.07363

Q14: Generate a boxplot with a box per genotype, what could you infer from the relative expression value between A/A and G/G displayed in this plot? Does the SNP effect the expression of ORMDL3?

```
library(ggplot2)

ggplot(expr) + aes(geno, exp, fill=geno) +
  geom_boxplot(notch=TRUE)
```



The SNP rs8067378 is associated with ORM DL3 expression, with A/A showing the highest expression and G/G showing the lowest. A allele is associated with higher ORM DL3 expression, while the G allele is associated with lower ORM DL3 expression.